Gebze Technical University 2023 FALL ELEC334 Microprocessors by Dr. Ihsan Cicek

Homework 1 Assigned: 25/10/2023 Due: 8/11/2023 17:00

1. (50 pts) The thermostat of Dr. Cicek's room heater had failed; he wants to replace the good old (and dead) electromechanical thermostat with a digital one. In his junkbox, he found one analog temperature sensor (10K PTC Resistive), one 10K potentiometer (to set the desired temperature) and two relays that can be operated by 3.3V logic outputs. He has endless supplies of resistors, capacitors, LEDs etc. Room heater has a 1kW resistive element for heating that is powered by 220V AC, and it also has a fan that is powered by 220V AC for cooling. He thinks that those two relays can be used to switch these on and off from the STM32G031 board he owns. Also, he thinks that some LED indicators would be nice to provide feedback about the status of the heater. However, due to his overloaded schedule he needs some assistance from the ELEC334 students for both hardware and software design. (You can use any drawing diagram software to implement your hardware and software diagrams including but not limited to draw.io).

a. (20 pts) Draw a block diagram based schematic design that uses STM32G031 board and mark the VDD, GND, reset, clock, analog input pins (to ADC) for the temperature sensor using a resistive voltage divider using a constant 10K resistor and sensor as well as the temperature control input using the potentiometer, and output pins for status LEDs (RED for heating, GREEN for cooling, include resistors for 5mA current and 3.3V VDD), for relays, so that a hardware design that will replace the failed thermostat can be built. Show all your work and necessary parts etc.

b. (30 pts) Draw a flow chart to guide the software implementation of an on - off based thermostat algorithm, that should perform the following tasks upon system reset:

1. Initialize the hardware (All LEDs are off, Heater off, Fan off)
2. Read and convert the ADC value for the temperature setting input
3. Read and convert the ADC value for the current ambient temperature
4. Compare [2] with [3]
5. If ambient temperature is lower than the setting turns the heater on, RED LED on, GREEN LED off, fan off, wait for 5 minutes, go to step 2
6. If ambient temperature is greater or equal to the set temperature turns the heater off, RED LED off, GREEN LED on, fan on, wait for 5 minutes, go to step 2

2. (25 pts) Let the value of the register r0 be 0x12345678. What are the values of the destination registers, the zero flag Z, and the negative flag N after the following ARM instructions?

a. (5 pts) MOVS R1, R0, LSL #4

b. (5 pts) MOVS R2, R0, LSR #8

c. (5 pts) MOVS R3, R0, ASR #8

d. (5 pts) MOVS R4, R0, ROR #4

e. (5 pts) MOVS R5, R0, LSL #29

3. (9 pts) Find the contents of register R2 after the following instructions run, assuming that memory contains the values shown on the right. Register R1 contains 0x24, and the memory system is little-endian.

Address	Contents
0x24	0x06
0x25	0xFC
0x26	0x03
0x27	0xFF

a. (3 pts) LDRSB R2, [R1]

b. (3 pts) LDRSH R2, [R1]

c. (3 pts) LDR R2, [R1]

4. (16 pts) Using R6 as a temporary storage for your code, write the assembly instruction(s) to perform the following operations on the ARM Cortex M0+:

a. (4 pts) Add 0xEA to register R1, placing the sum in register R0.

b. (4 pts) Subtract 0xDEADBEEF from register R2, placing the result in register R2.

c. (4 pts) Add the value 0xE1EC0334 to register R4, placing the sum in register R3.

d. (4 pts) Place -334 into register R4 using the two's complement representation.

Please deliver your solutions in printed form to the teaching assistants of the course (Not to the Instructor) before the designated deadline. Late submissions will not be accepted and automatically graded with 0.